CRC Lab — Calculation Report

Generated by CRC Lab

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1 Inputs

Message	`HELLO`
Data bits	`0100100001000101010011000100110001001111` (40 bits)
Generator	`1011` (4 bits)
Appended zeros	3

2 Sender: polynomial long division

Augmented dividend (data + 3 zeros):

`0100100001000101010011000100110001001111000`

Generator: `1011`. At each step a leading 1 is cancelled by XOR with the generator (⊕ marks such a step):

step	xor	register after step
0	—	01001000010001010100110001001100010011111000
1	⊕	00010000010001010100110001001100010011111000
2	—	00010000010001010100110001001100010011111000
3	⊕	00000110010001010100110001001100010011111000
4	—	00000110010001010100110001001100010011111000
5	⊕	00000011110001010100110001001100010011111000
6	⊕	00000001000001010100110001001100010011111000
7	⊕	00000000011001010100110001001100010011111000
8	—	00000000011001010100110001001100010011111000
9	⊕	00000000001111010100110001001100010011111000
10	⊕	00000000000010001010011000100110001001111000
11	⊕	00000000000001110100110001001100010011111000
12	—	00000000000001110100110001001100010011111000
13	⊕	00000000000000101100110001001100010011111000
14	⊕	00000000000000000000110001001100010011111000
15	—	00000000000000000000110001001100010011111000
16	—	00000000000000000000110001001100010011111000
17	—	00000000000000000000110001001100010011111000
18	—	00000000000000000000110001001100010011111000
19	—	00000000000000000000110001001100010011111000
20	⊕	00000000000000000000011101001100010011111000
21	⊕	00000000000000000000001011001100010011111000
22	⊕	00000000000000000000000000000110001001111000
23	—	00000000000000000000000000000110001001111000
24	—	00000000000000000000000000000110001001111000
25	—	00000000000000000000000000000110001001111000
26	—	00000000000000000000000000000110001001111000
27	—	00000000000000000000000000000110001001111000
28	⊕	00000000000000000000000000000011101001111000
29	⊕	00000000000000000000000000000001011001111000
30	⊕	00000000000000000000000000000000000001111000
31	—	00000000000000000000000000000000000001111000
32	—	00000000000000000000000000000000000001111000
33	—	00000000000000000000000000000000000001111000
34	—	00000000000000000000000000000000000001111000
35	—	00000000000000000000000000000000000001111000
36	⊕	00000000000000000000000000000000000100000
37	⊕	00000000000000000000000000000000000001100
38	—	00000000000000000000000000000000000001100
39	⊕	00000000000000000000000000000000000000111

3 Transmitted frame

CRC remainder: `111`. Frame = data + CRC:

`01001000010001010100110001001100010011111111` (43 bits)

4 Receiver verification

Received frame: `01001000010001010100110001001100010011111111`. Flipped bit positions: none. Dividing by `1011` gives remainder `000`.

VALID — no error

Remainder is all zero, so the frame arrived intact.